

Sampling of Univariate Random Variables

Special Mathematics Lectures – Introduction to probability

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The main goal of this report is to introduce some methods for generating (pseudo)random numbers from a given probability distribution. The main inspiration of this report is Section 4.11 in [1].

1 Uniform distribution

In this section, we construct a random variable that will be useful for later sections. Firstly, let us construct a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where the sample space Ω is the closed interval $[0, 1]$.

The σ -algebra $\mathcal{F}$ is required to contain all open sets in $[0, 1]$. For $a, b, c, d \in (0, 1)$, the subsets $[0, a)$, (b, c) , and $(d, 1]$ are open in $[0, 1]$. Any subset of $[0, 1]$ which can be expressed as an arbitrary union of these sets is also open in $[0, 1]$ and thus belongs to $\mathcal{F}$. By the definition of σ -algebras (see Definition 1.1.2 in [2]), $\mathcal{F}$ also contains the complements of these sets which are closed in $[0, 1]$. Therefore, $\mathcal{F}$, called **Borel σ -algebra**, contains all open and closed subsets of $[0, 1]$.

The next step now is to define a probability measure $\mathbb{P}$ on the measurable space $(\Omega, \mathcal{F})$. For $a, b \in [0, 1]$ such that $a < b$, the probability measure $\mathbb{P}$ is given by

$$\mathbb{P}([a, b]) = \mathbb{P}([a, b)) = \mathbb{P}((a, b]) = \mathbb{P}((a, b)) = b - a. \quad (1)$$

More generally, for any set $A \in \mathcal{F}$, we define a *covering* of A as a countable collection $\{I_n\}_{n=1}^{\infty}$ of closed intervals such that $A \subset \bigcup_{n=1}^{\infty} I_n$. Then, the probability of A is defined by

$$\mathbb{P}(A) = \inf \left\{ \sum_{n=1}^{\infty} \mathbb{P}(I_n) \mid \{I_n\}_{n=1}^{\infty} \text{ is a covering of } A \right\}. \quad (2)$$

Finally, we define a random variable $X : \Omega \rightarrow \mathbb{R}$ by $X(\omega) = \omega$ for all $\omega \in \Omega$. This random variable is simply the identity function $\text{id}_{[0,1]}$, which is always nice. Indeed, we have

$$X^{-1}((-\infty, x]) = \begin{cases} \emptyset, & x < 0, \\ [0, x], & 0 \leq x < 1, \\ [0, 1], & x \geq 1, \end{cases} \quad (3)$$

which belongs to $\mathcal{F}$ for all $x \in \mathbb{R}$. Therefore, X is a random variable as claimed. X is said to obey the *uniform distribution* $U(0, 1)$, which is denoted by $X \sim U(0, 1)$. In addition, the cumulative distribution function (cdf) $F_X : \mathbb{R} \rightarrow [0, 1]$ of X is given by

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(X^{-1}((-\infty, x])) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases} \quad (4)$$

The random variable X is a continuous random variable since the cdf of X is absolutely continuous and it can be expressed as

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \quad (5)$$

where the probability density function (pdf) $f_X : \mathbb{R} \rightarrow \mathbb{R}$ of X is given by

$$f_X(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

In practice, to approximate the uniform distribution $U(0, 1)$ on computers, people use *pseudo-random numbers* which are computed by a certain formula. A sequence of pseudo-random numbers computed by using this formula has a chaotic property, which makes the sequence unpredictable. However, pseudo-random numbers are not truly random and the problem of generating random numbers on computers is still an open problem. Nevertheless, we shall suppose that these numbers are random in this report.

2 Inverse transform method

In this section, we consider a uniformly distributed random variable $U \sim U(0, 1)$ and a known cdf $F : \mathbb{R} \rightarrow [0, 1]$ satisfying certain conditions: F is increasing and right-continuous, $\lim_{x \rightarrow -\infty} F(x) = 0$, and $\lim_{x \rightarrow \infty} F(x) = 1$. We want to construct a random variable X with cdf F from U .

Theorem 1. (a) If F is piecewise constant with countably many discontinuities at $x_1, x_2, \dots$, the random variable $X = g(U)$ where the function $g : [0, 1] \rightarrow \mathbb{R}$ is given by

$$g(u) = \begin{cases} x_1, & 0 \leq u \leq F(x_1), \\ x_k, & F(x_{k-1}) < u \leq F(x_k), \quad k \geq 2, \end{cases} \quad (7)$$

is a discrete random variable with a cumulative distribution function F .

(b) If F is a strictly increasing and absolutely continuous function, the random variable $X = F^{-1}(U)$ has a cumulative distribution function F .

Proof. (a) It is clear that the range of X is countable since $\text{Ran}(X) = \{x_1, x_2, \dots\}$. Hence, X is a discrete random variable. In addition, the pmf of X is given by

$$\pi_1 = \mathbb{P}(X = x_1) = \mathbb{P}(U \leq F(x_1)) = F(x_1), \quad (8)$$

$$\pi_k = \mathbb{P}(X = x_k) = \mathbb{P}(F(x_{k-1}) < U \leq F(x_k)) = F(x_k) - F(x_{k-1}), \quad k \geq 2. \quad (9)$$

We can easily verify that

$$F_X(x) = \mathbb{P}(X \leq x) = \sum_{k=1}^{k_m} \mathbb{P}(X = x_k) = \sum_{k=1}^{k_m} \pi_k = F(x), \quad (10)$$

where k_m is the largest index such that $x_{k_m} \leq x$.

(b) The cdf of X is given by

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x), \quad (11)$$

which is the desired result. $\square$

Suppose that we generate n independent random numbers, which are *realizations* of n independent and identically distributed (i.i.d.) random variables $X_1, X_2, \dots, X_n$ with a common distribution function F . If these random variables are discrete, it is natural to count the number of times that the generated random numbers take a particular value x_k . Otherwise, if these random variables are continuous, we can count the number of times that the generated random numbers fall into a particular interval $[a, b]$. In either case, we obtain the number of occurrences of a particular event A with a certain probability $\mathbb{P}(A) = p$. More precisely, suppose $A_1, A_2, \dots, A_n$ are n independent events such that $\mathbb{P}(A_i) = p$ for all $i = 1, 2, \dots, n$.

$$A_i = \begin{cases} \{X_i = x_k\}, & \text{if discrete,} \\ \{X_i \in [a, b]\}, & \text{if continuous.} \end{cases} \quad (12)$$

Then, define the random variable $N_A = \sum_{i=1}^n \mathbf{1}_{A_i}$, where $\mathbf{1}_{A_i}$ is the indicator function of the event A_i . A realization n_A of N_A counting the number of occurrences of the event A in our experiment is called the *frequency* of the event A . If the number of generated random numbers n is sufficiently large, we expect that the ratio $\frac{n_A}{n}$, also called *relative frequency*, is close to the probability p of the event A . The following theorem justifies this intuition.

Theorem 2 (The law of averages). *For every $\epsilon > 0$, we have*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{N_A}{n} - p \right| \leq \epsilon \right) = 1. \quad (13)$$

Remark. This statement says that the random variable $\frac{N_A}{n}$ converges *in probability* to the random variable $p\mathbf{1}_\Omega$ where $\mathbf{1}_\Omega(\omega) = 1$ for all elements ω in the sample space Ω . The subject of convergence of random variables will be covered in the later part of the course.

The proof of this theorem is provided in the Appendix. In the following examples, we apply Theorem 1 to construct random variables with various distributions from a uniformly distributed random variable $U \sim U(0, 1)$ and use Theorem 2 to check that the generated random numbers indeed follow the desired distributions. This can be done by generating a large *sample*, which is a collection of realizations of random variables, and plotting the histogram showing relative frequencies of outcomes in the sample. From these plots, we expect to recover the pmf for discrete random variables and the pdf for continuous random variables.

Example 3 (Bernoulli and Binomial distributions). Recall that the *Bernoulli distribution* with parameter $p \in [0, 1]$ is a discrete distribution describing a single experiment that results in success ($X = 1$) with probability p and failure ($X = 0$) with probability $1 - p$. Its pmf is given by

$$\pi_k = \begin{cases} 1 - p, & k = 0, \\ p, & k = 1. \end{cases} \quad (14)$$

The cdf of this distribution is

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases} \quad (15)$$

By Theorem 1(a), if $U \sim U(0, 1)$, the random variable with a Bernoulli distribution is given by

$$X = \chi_{(1-p, 1]}(U) \quad (16)$$

where $\chi_{(1-p, 1]} : [0, 1] \rightarrow \mathbb{R}$ is the characteristic function of the interval $(1 - p, 1]$ defined by

$$\chi_{(1-p, 1]}(u) = \begin{cases} 0, & 0 \leq u \leq 1 - p, \\ 1, & 1 - p < u \leq 1. \end{cases} \quad (17)$$

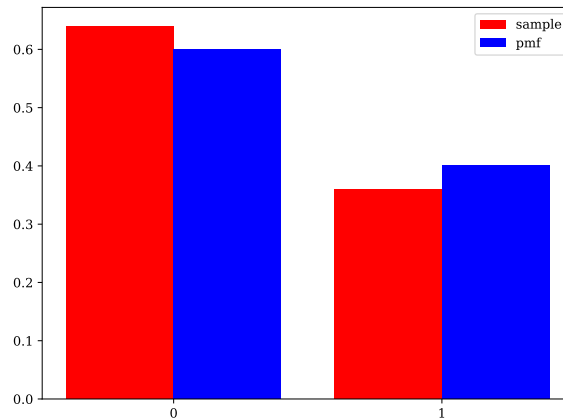


Figure 1: The graph shows a sample of 500 random numbers obeying the Bernoulli distribution with $p = 0.4$.

The *Binomial distribution* with parameters $n \in \mathbb{N}$ and $p \in (0, 1)$ represents the number of successes in n independent Bernoulli trials. Its pmf is given by

$$\pi_k = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n. \quad (18)$$

Consider n i.i.d. Bernoulli random variables $X_1, X_2, \dots, X_n$ with parameter p . These random variables can be obtained from n i.i.d. uniformly distributed random variables $U_1, U_2, \dots, U_n \sim U(0, 1)$ by using (16). A random variable Y following the binomial distribution with n trials and success probability p can be constructed as the sum of these Bernoulli random variables (see Example 2.5.10. in [2]), i.e.,

$$Y = \sum_{i=1}^n X_i = \sum_{i=1}^n \chi_{(1-p, 1]}(U_i). \quad (19)$$

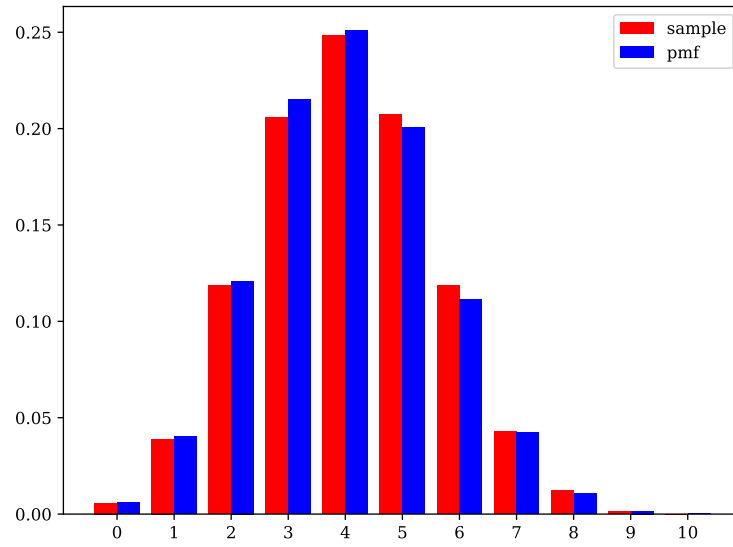


Figure 2: The graph shows a sample of 10,000 random numbers obeying the Binomial distribution with $n = 10$ and $p = 0.4$.

Example 4 (Geometric distribution). Recall that the *Geometric distribution* with parameter $p \in (0, 1]$ has a pmf given by

$$\pi_k = p(1-p)^{k-1}, \quad k = 1, 2, \dots \quad (20)$$

The cdf of this distribution is given by

$$F(x) = \sum_{k=1}^{\lfloor x \rfloor} \pi_k = \sum_{k=1}^{\lfloor x \rfloor} p(1-p)^{k-1} = p \frac{1 - (1-p)^{\lfloor x \rfloor}}{1 - (1-p)} = 1 - (1-p)^{\lfloor x \rfloor}, \quad (21)$$

where $\lfloor x \rfloor$ denotes the largest integer less than or equal to x . Now, consider the inequality $u \leq F(n)$, which can be rewritten as

$$u \leq 1 - (1-p)^n \iff n \ln(1-p) \leq \ln(1-u) \iff n \geq \frac{\ln(1-u)}{\ln(1-p)}. \quad (22)$$

Similarly, the inequality $u > F(n-1)$ is equivalent to $n < 1 + \frac{\ln(1-u)}{\ln(1-p)}$. Therefore, we can define the function $g : [0, 1] \rightarrow \mathbb{R}$ by

$$g(u) = \left\lceil \frac{\ln(1-u)}{\ln(1-p)} \right\rceil, \quad (23)$$

where $\lceil x \rceil$ denotes the smallest integer greater than or equal to x . By Theorem 1(a), if $U \sim U(0, 1)$, the random variable $X = g(U)$ has a geometric distribution with parameter p .

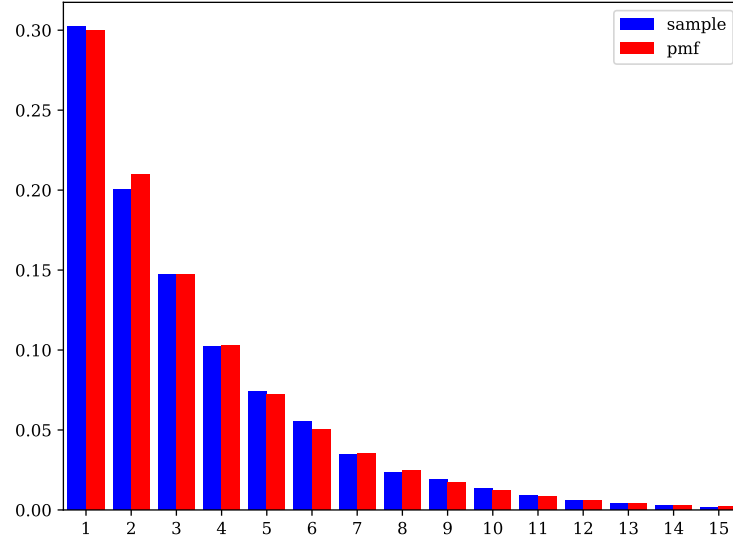


Figure 3: The graph shows a sample of 10,000 random numbers obeying the geometric distribution with $p = 0.3$.

Example 5 (Exponential distribution). The *exponential distribution* with parameter $\lambda > 0$, denoted by $\text{Exp}(\lambda)$, has a pdf given by

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad (24)$$

and $f(x) = 0$ for $x < 0$. The cdf of this distribution is given by

$$F(x) = \int_{-\infty}^x f(t) dt = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases} \quad (25)$$

If we restrict the domain of F to $[0, \infty)$, then the inverse function $F^{-1} : [0, 1] \rightarrow [0, \infty)$ is given by

$$F^{-1}(y) = -\frac{1}{\lambda} \ln(1 - y). \quad (26)$$

By Theorem 1(b), if $U \sim U(0, 1)$, the random variable

$$X = F^{-1}(U) = -\frac{1}{\lambda} \ln(1 - U) \quad (27)$$

has an exponential distribution with parameter λ .

The exponential distribution arises as the continuum limit of the geometric distribution as the parameter p goes to zero. Consider a random variable N following the geometric distribution with a success probability p . Suppose that each trial in the geometric distribution takes a small amount of time Δx and the success probability p is proportional to Δx , i.e. $p = \lambda \Delta x$. From (21), the probability that the first success occurs at time less than x is approximately given by

$$\mathbb{P}(N\Delta x \leq x) \approx 1 - (1 - \lambda\Delta x)^{x/\Delta x} \xrightarrow{\Delta x \rightarrow 0} 1 - (e^{-\lambda})^x = 1 - e^{-\lambda x}. \quad (28)$$

Hence, $X = N\Delta x$ as $\Delta x \rightarrow 0$ follows the exponential distribution with parameter λ . That is, the random variable X describes the waiting time until the first success. The parameter λ is the inverse of the average waiting time since

$$\mathbb{E}[X] = \int_0^{\infty} x \lambda e^{-\lambda x} dx = -x e^{-\lambda x} \Big|_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx = -\frac{e^{-\lambda x}}{\lambda} \Big|_0^{\infty} = \frac{1}{\lambda}. \quad (29)$$

This probability distribution is widely used in physics to describe radioactive or particle decay. In this context, λ is also called *decay width* and $\tau = 1/\lambda$ is the *mean lifetime*.

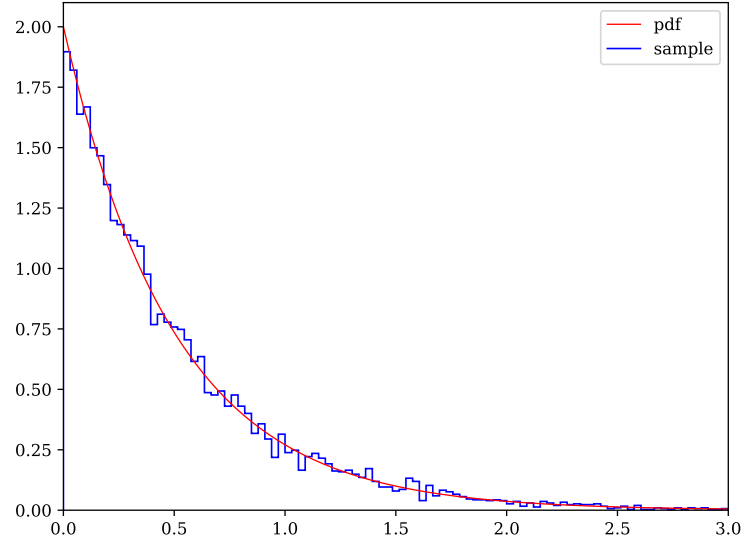


Figure 4: The graph shows a sample of 10,000 random numbers obeying the exponential distribution with $\lambda = 2$.

Example 6 (Cauchy distribution). Consider a random variable following the *Cauchy distribution* with a pdf given by

$$f(x) = \frac{1}{\pi} \frac{\gamma}{(x - x_0)^2 + \gamma^2}, \quad (30)$$

where $x_0 \in \mathbb{R}$ and $\gamma > 0$ are parameters. The cdf of this distribution is given by

$$F(x) = \int_{-\infty}^x f(t) dt = \frac{1}{\pi} \int_{-\infty}^x \frac{\gamma}{(t - x_0)^2 + \gamma^2} dt = \frac{1}{2} + \frac{1}{\pi} \arctan \left(\frac{x - x_0}{\gamma} \right). \quad (31)$$

This function can be easily inverted to

$$F^{-1}(y) = x_0 + \gamma \tan \left[\left(y - \frac{1}{2} \right) \pi \right]. \quad (32)$$

By Theorem 1(b), if $U \sim U(0, 1)$, the random variable

$$X = F^{-1}(U) = x_0 + \gamma \tan \left[\left(U - \frac{1}{2} \right) \pi \right] \quad (33)$$

has a Cauchy distribution with parameters x_0 and γ .

In probability theory, the Cauchy distribution is often used as an example of a continuous random variable that has an undefined expectation since the following integral does not converge:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x) dx = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\gamma x}{(x - x_0)^2 + \gamma^2} dx \quad (34)$$

$$= \frac{\gamma}{\pi} \int_{-\infty}^{\infty} \frac{y + x_0}{y^2 + \gamma^2} dy \quad (35)$$

$$= \frac{\gamma}{\pi} \left[\frac{1}{2} \ln(y^2 + \gamma^2) + \frac{x_0}{\gamma} \arctan \left(\frac{y}{\gamma} \right) \right] \Bigg|_{-\infty}^{\infty}. \quad (36)$$

Furthermore, the Cauchy distribution commonly appears under the name of the “Breit-Wigner distribution” in the scattering theory of non-relativistic quantum mechanics. In this context, x_0 is the energy of the resonance and γ is the half-width.

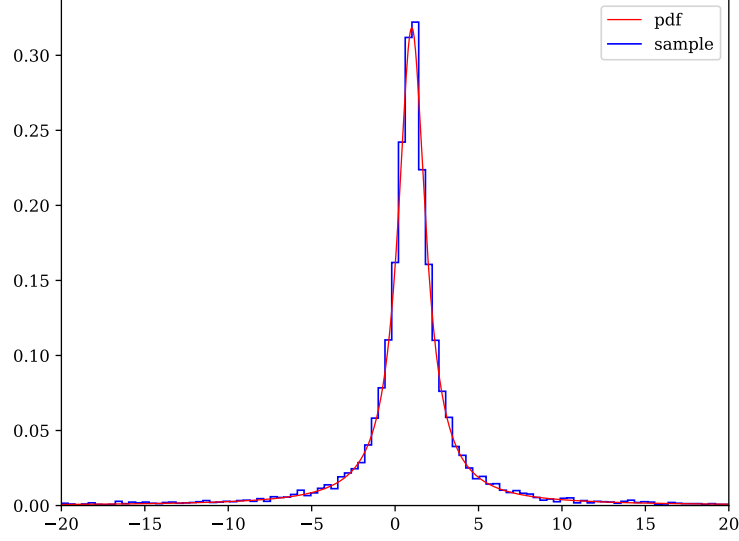


Figure 5: The graph shows a sample of 10,000 random numbers obeying the Cauchy distribution with $x_0 = 1$ and $\gamma = 1$.

3 Rejection method

In this section, we introduce a method for generating random numbers following a continuous distribution given its density function f . Despite the simplicity of the inverse transform method, it is not always possible to find an explicit formula for the inverse function F^{-1} of the cdf F . It would be easier if we can directly use the pdf f to generate the desired distribution. In the *rejection method*, we require an additional continuous random variable Y which is independent of U and has a pdf f_Y such that $f(x) \leq a f_Y(x)$ for all $x \in \mathbb{R}$ and some constant $a \geq 1$. Define an event $E = \{aU f_Y(Y) \leq f(Y)\}$. We want to calculate the conditional probability

$$\mathbb{P}(Y \leq x \mid E) = \frac{\mathbb{P}(\{Y \leq x\} \cap E)}{\mathbb{P}(E)}. \quad (37)$$

By the partition theorem for probability (see Theorem 1.2.6 in [2]), we have

$$\mathbb{P}(E) = \int_{-\infty}^{\infty} \mathbb{P}(E \mid Y = y) f_Y(y) dy = \int_{-\infty}^{\infty} \mathbb{P}\left(aU f_Y(Y) \leq f(Y) \mid Y = y\right) f_Y(y) dy \quad (38)$$

$$= \int_{-\infty}^{\infty} \mathbb{P}\left(U \leq \frac{f(y)}{a f_Y(y)}\right) f_Y(y) dy \quad (39)$$

$$= \int_{-\infty}^{\infty} \frac{f(y)}{a f_Y(y)} f_Y(y) dy \quad (40)$$

$$= \frac{1}{a} \int_{-\infty}^{\infty} f(y) dy = \frac{1}{a}. \quad (41)$$

Notice that $\mathbb{P}(\{Y \leq x\} \cap E \mid Y = y) = 0$ for $y > x$. Hence, by similar calculations, we have

$$\mathbb{P}(\{Y \leq x\} \cap E) = \int_{-\infty}^{\infty} \mathbb{P}(\{Y \leq x\} \cap E \mid Y = y) f_Y(y) dy \quad (42)$$

$$= \int_{-\infty}^x \mathbb{P}(E \mid Y = y) f_Y(y) dy \quad (43)$$

$$= \frac{1}{a} \int_{-\infty}^x f(y) dy = \frac{1}{a} F(x). \quad (44)$$

As a result, we obtain

$$\mathbb{P}(Y \leq x \mid E) = \frac{\mathbb{P}(\{Y \leq x\} \cap E)}{\mathbb{P}(E)} = F(x). \quad (45)$$

This equation tells us that the random variable Y behaves as if it has a distribution function F given that the event E occurs. Hence, the rejection method can be implemented as follows: we generate a realization (y, u) of the random vector (Y, U) and check if the event E occurs. If E occurs, we accept y as a realization of the desired distribution. Otherwise, we reject y and repeat the process until we get a realization that is accepted. Since $\mathbb{P}(E) = 1/a$, the process of generating a realization of the desired distribution takes a shorter time if the constant a is closer to 1. Therefore, the choice of the random variable Y is crucial for the efficiency of the algorithm.

Example 7 (Gamma distribution). The *Gamma distribution* with parameters $\alpha > 0$ and $\beta > 0$, also denoted by $\Gamma(\alpha, \beta)$, has a pdf given by

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x \geq 0, \quad (46)$$

where $\Gamma(\alpha)$ is the Gamma function defined by

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt. \quad (47)$$

For $\alpha = 1$ and $\beta = \lambda$, the Gamma distribution $\Gamma(\alpha, \beta)$ reduces to the exponential distribution $\text{Exp}(\lambda)$. For $\beta = \frac{1}{2}$ and $\alpha = \frac{d}{2}$ for some integer $d \geq 1$, the Gamma distribution $\Gamma(\alpha, \beta)$ becomes the *chi-squared distribution* $\chi^2(d)$ with d degrees of freedom, which is important in statistics.

In this example, we consider only the case $\alpha > 1$ and apply the rejection method using $Y \sim \text{Exp}(\lambda)$. To do this, we need to find an upper bound a of the function $g : [0, \infty) \rightarrow \mathbb{R}$ given by

$$g(x) = \frac{f(x)}{f_Y(x)} = \frac{\beta^\alpha}{\lambda \Gamma(\alpha)} x^{\alpha-1} e^{(\lambda-\beta)x}. \quad (48)$$

Supposing that $\lambda < \beta$, we have $\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow \infty} g(x) = 0$. Since $g(x)$ is non-negative, there exists a global maximum of g at some point $x_m > 0$. To find x_m , we can solve the equation $g'(x_m) = 0$, which is equivalent to

$$(\alpha - 1)x_m^{\alpha-2} e^{(\lambda-\beta)x_m} + (\lambda - \beta)x_m^{\alpha-1} e^{(\lambda-\beta)x_m} = 0 \iff x_m = \frac{\alpha - 1}{\beta - \lambda}. \quad (49)$$

Hence, the least upper bound of g is given by

$$g(x_m) = \frac{\beta^\alpha}{\lambda \Gamma(\alpha)} \left(\frac{\alpha - 1}{\beta - \lambda} \right)^{\alpha-1} e^{1-\alpha}. \quad (50)$$

This bound can be lowered by choosing a suitable value of λ . This can be done by maximizing the quantity $h(\lambda) = \lambda(\beta - \lambda)^{\alpha-1}$ for λ such that $0 < \lambda < \beta$. Since $h(0) = h(\beta) = 0$ and $h(\lambda) \geq 0$, there exists a global maximum λ_m satisfying $h'(\lambda_m) = 0$, which is equivalent to

$$(\beta - \lambda_m)^{\alpha-1} - \lambda_m(\alpha - 1)(\beta - \lambda_m)^{\alpha-2} = 0 \iff \lambda_m = \frac{\beta}{\alpha}. \quad (51)$$

As a result, by using $\lambda = \lambda_m = \beta/\alpha$, we can choose a such that

$$a = g(x_m) = \frac{\alpha^\alpha}{\Gamma(\alpha)} e^{1-\alpha}. \quad (52)$$

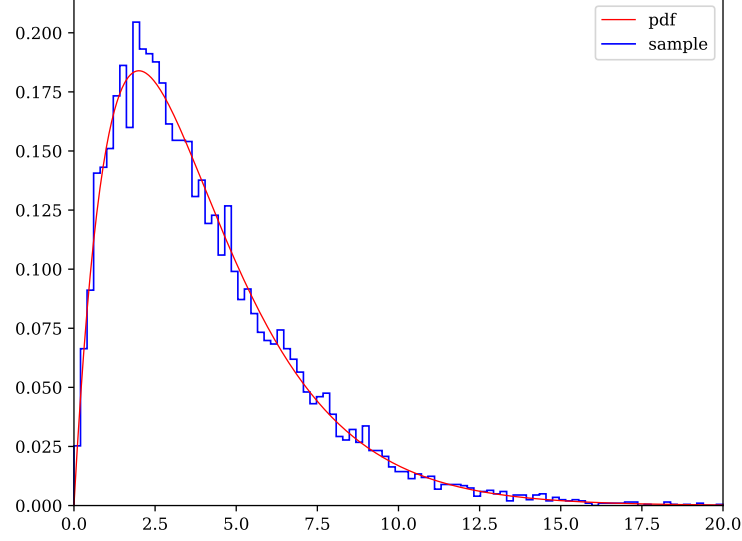


Figure 6: The graph shows a sample of 10,000 random numbers obeying the gamma distribution with $\alpha = 2$ and $\beta = 0.5$.

4 Box-Muller method

An important distribution in probability theory and statistics is the *normal distribution*, denoted by $N(\mu, \sigma^2)$, which has a pdf given by

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}, \quad (53)$$

for $\mu \in \mathbb{R}$ and $\sigma > 0$. The two parameters μ and σ^2 are the expectation and variance, i.e. for $X \sim N(\mu, \sigma^2)$, we can show that $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$. From a random variable $Z \sim N(0, 1)$, we obtain a random variable $X \sim N(\mu, \sigma^2)$ by the transformation $X = \mu + \sigma Z$. Indeed, the pdf of Z is given by

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}. \quad (54)$$

By setting $z = \frac{x-\mu}{\sigma}$, the pdf of X is given by

$$f_X(x) = f_Z(z) \frac{dz}{dx} = f_Z\left(\frac{x-\mu}{\sigma}\right) \frac{1}{\sigma} = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}. \quad (55)$$

Therefore, it suffices to focus on the distribution $N(0, 1)$. An efficient method for doing this is the *Box-Muller method*, which uses two i.i.d. uniformly distributed random variables $U_1, U_2 \sim U(0, 1)$.

Theorem 8 (Box-Muller method). *Given two i.i.d. random variables $U_1, U_2 \sim U(0, 1)$, two i.i.d. random variables $Z_1, Z_2 \sim N(0, 1)$ can be obtained by the following transformation:*

$$R = \sqrt{-2\ln(U_1)}, \quad V = 2\pi U_2, \quad (56)$$

$$Z_1 = R \cos(V), \quad Z_2 = R \sin(V). \quad (57)$$

Proof. Let Z_1, Z_2 be independent random variables and $Z_1, Z_2 \sim N(0, 1)$. Their joint pdf is

$$f_{Z_1, Z_2}(z_1, z_2) = f_{Z_1}(z_1) f_{Z_2}(z_2) = \frac{1}{\sqrt{2\pi}} e^{-z_1^2/2} \frac{1}{\sqrt{2\pi}} e^{-z_2^2/2} = \frac{1}{2\pi} e^{-(z_1^2 + z_2^2)/2}. \quad (58)$$

Now, we express the random vector (Z_1, Z_2) in polar coordinates, i.e.

$$(Z_1, Z_2) = (R \cos(V), R \sin(V)), \quad (59)$$

where R, V are some random variables with $\text{Ran}(R) = (0, +\infty)$ and $\text{Ran}(V) = (0, 2\pi)$. Consider a diffeomorphism $\varphi : (0, +\infty) \times (0, 2\pi) \rightarrow \mathbb{R}^2 \setminus \{0\}$ defined by $\varphi(r, v) = (r \cos(v), r \sin(v))$. Then, the joint pdf of the random vector (R, V) is

$$f_{R,V}(r, v) = f_{Z_1, Z_2}(\varphi(r, v)) |\det(\mathcal{J}_\varphi(r, v))| \quad (60)$$

$$= \frac{1}{2\pi} e^{-r^2/2} \left| \det \begin{pmatrix} \cos(v) & -r \sin(v) \\ \sin(v) & r \cos(v) \end{pmatrix} \right| \quad (61)$$

$$= \frac{1}{2\pi} r e^{-r^2/2}. \quad (62)$$

The corresponding marginal pdfs are

$$f_R(r) = \int_0^{2\pi} f_{R,V}(r, v) dv = \int_0^{2\pi} \frac{1}{2\pi} r e^{-r^2/2} dv = r e^{-r^2/2}, \quad (63)$$

$$f_V(v) = \int_0^\infty f_{R,V}(r, v) dr = \int_0^\infty \frac{1}{2\pi} r e^{-r^2/2} dr = -\frac{1}{2\pi} e^{-r^2/2} \Big|_{r=0}^{r=\infty} = \frac{1}{2\pi}. \quad (64)$$

Hence, $f_{R,V}(r, v) = f_R(r)f_V(v)$, which implies that R and V are independent. Now, consider the transformation $R = \sqrt{-2 \ln(U_1)}$ and $V = 2\pi U_2$, where $\text{Ran}(U_1) = \text{Ran}(U_2) = (0, 1)$. Then, the pdfs of U_1 and U_2 are given by

$$f_{U_1}(u_1) = f_R(\sqrt{-2 \ln(u_1)}) \left| \frac{d}{du_1} (\sqrt{-2 \ln(u_1)}) \right| = \sqrt{-2 \ln(u_1)} \cdot u_1 \cdot \frac{1/u_1}{\sqrt{-2 \ln(u_1)}} = 1,$$

$$f_{U_2}(u_2) = f_V(2\pi u_2) \left| \frac{d}{du_2} (2\pi u_2) \right| = \frac{1}{2\pi} 2\pi = 1.$$

Thus, $U_1, U_2 \sim U(0, 1)$ and they are also independent because R and V are independent. Since the transformation from (Z_1, Z_2) to (U_1, U_2) is invertible, we also obtain independent random variables $Z_1, Z_2 \sim N(0, 1)$ given independent random variables $U_1, U_2 \sim U(0, 1)$ by following the whole calculations in the reverse direction. $\square$

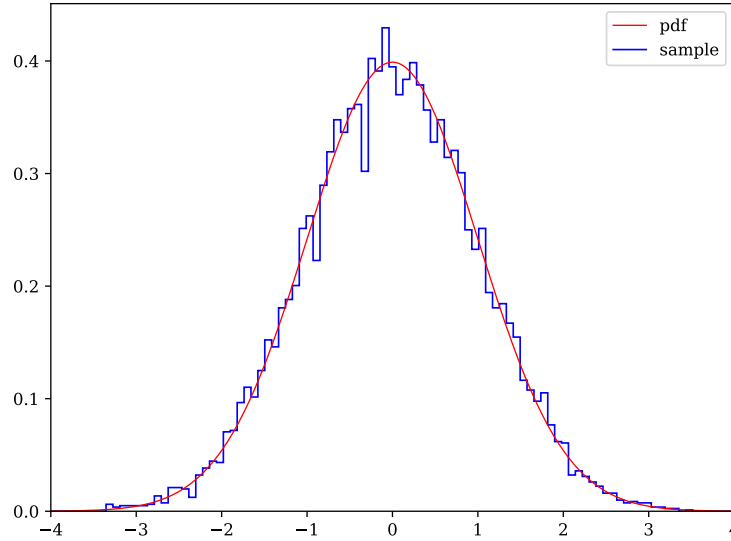


Figure 7: The graph shows a sample of 10,000 random numbers obeying the normal distribution with $\mu = 0$ and $\sigma = 1$.

5 Knuth's algorithm

One final distribution that we want to discuss is the *Poisson distribution* with parameter $\lambda > 0$, which has a pmf given by

$$\pi_k = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots \quad (65)$$

It is possible to use the inverse transform method for the Poisson distribution. However, it may not be efficient since there is no closed-form formula to calculate n such that $F(n-1) < u \leq F(n)$ for a given $u \in (0, 1)$. An alternative method due to [Donald Knuth](#) is to utilize i.i.d. random variables following the exponential distribution $\text{Exp}(\lambda)$. Suppose that $X_1, X_2, \dots$ are i.i.d. random variables with an exponential distribution $\text{Exp}(\lambda)$. For a given integer $n \geq 1$, define the random variable S_n by

$$S_n = \sum_{i=1}^n X_i. \quad (66)$$

We claim that the pdf of S_n is given by

$$f_{S_n}(s) = \frac{\lambda^n}{(n-1)!} s^{n-1} e^{-\lambda s}, \quad s \geq 0. \quad (67)$$

For $n = 1$, we simply have $S_1 = X_1$, which has the desired pdf. Assume that the claim is true for some $n \geq 1$. Observe that S_n and X_{n+1} are independent since X_{n+1} is independent of $X_1, \dots, X_n$. Then, using the convolution formula (Theorem 3.5.3 in [2]), we have

$$f_{S_{n+1}}(s) = f_{S_n+X_{n+1}}(s) \quad (68)$$

$$= \int_0^s f_{S_n}(t) f_{X_{n+1}}(s-t) dt = \int_0^s \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t} \cdot \lambda e^{-\lambda(s-t)} dt \quad (69)$$

$$= \frac{\lambda^{n+1}}{(n-1)!} e^{-\lambda s} \int_0^s t^{n-1} dt = \frac{\lambda^{n+1}}{n!} s^n e^{-\lambda s}. \quad (70)$$

Hence, by induction, the claim is true for all $n \geq 1$. Now, observe first that

$$\mathbb{P}(1 < S_1) = \mathbb{P}(1 < X_1) = \int_1^\infty \lambda e^{-\lambda s} ds = e^{-\lambda} = \pi_0. \quad (71)$$

For $n \geq 1$, we have

$$\mathbb{P}(S_n \leq 1 < S_{n+1}) = \mathbb{P}(S_n \leq 1 < S_n + X_{n+1}) \quad (72)$$

$$= \int_0^1 \mathbb{P}(S_n \leq 1 < S_n + X_{n+1} \mid S_n = s) f_{S_n}(s) ds \quad (73)$$

$$= \int_0^1 \mathbb{P}(X_{n+1} > 1-s) \frac{\lambda^n}{(n-1)!} s^{n-1} e^{-\lambda s} ds \quad (74)$$

$$= \int_0^1 \int_{1-s}^\infty \lambda e^{-\lambda x} dx \frac{\lambda^n}{(n-1)!} s^{n-1} e^{-\lambda s} ds \quad (75)$$

$$= \int_0^1 e^{-\lambda(1-s)} \frac{\lambda^n}{(n-1)!} s^{n-1} e^{-\lambda s} ds \quad (76)$$

$$= e^{-\lambda} \frac{\lambda^n}{(n-1)!} \int_0^1 s^{n-1} ds = e^{-\lambda} \frac{\lambda^n}{n!} = \pi_n. \quad (77)$$

Finally, denote the probability space where $S_1, S_2, \dots$ are defined by $(\Omega_P, \mathcal{F}_P, \mathbb{P}_P)$. For every $\omega \in \Omega_P$, we have an increasing sequence $S_1(\omega), S_2(\omega), \dots$. Consequently, if we define a discrete random variable N by

$$N(\omega) = \min\{n \geq 0 \mid 1 < S_{n+1}(\omega)\}, \quad \omega \in \Omega_P, \quad (78)$$

then N follows the Poisson distribution with parameter λ . Note that the condition $S_n(\omega) \leq 1$ is already satisfied by the operation of taking the minimum.

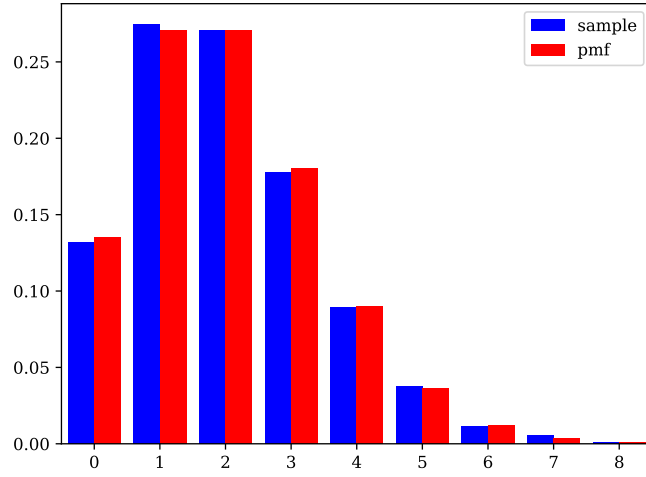


Figure 8: The graph shows a sample of 10,000 random numbers obeying the Poisson distribution with $\lambda = 2$.

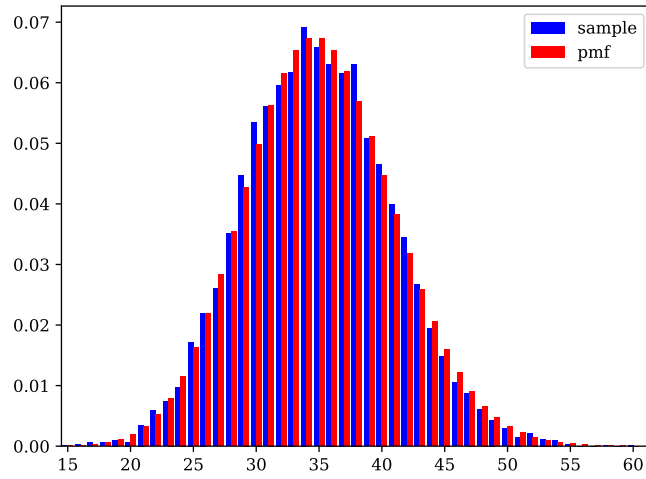


Figure 9: The graph shows a sample of 10,000 random numbers obeying the Poisson distribution with $\lambda = 35$.

Appendices

Theorem 9 (The law of averages). *For every $\epsilon > 0$, we have*

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\left| \frac{N_A}{n} - p \right| \leq \epsilon \right) = 1. \quad (79)$$

Proof. Observe that N_A follows the binomial distribution with parameters n and p . Hence, we have

$$\mathbb{P} \left(\frac{N_A}{n} > p + \epsilon \right) = \sum_{k > n(p+\epsilon)} \mathbb{P}(N_A = k) = \sum_{k > n(p+\epsilon)} \binom{n}{k} p^k (1-p)^{n-k}. \quad (80)$$

Let λ be an arbitrary positive number. Observe that $e^{\lambda(k-n(p+\epsilon))} \geq 1$ for every integer k such that $k > n(p+\epsilon)$. Hence, letting $q = 1 - p$, we have

$$\mathbb{P}\left(\frac{N_A}{n} > p + \epsilon\right) \leq \sum_{k > n(p+\epsilon)} e^{\lambda(k-n(p+\epsilon))} \binom{n}{k} p^k q^{n-k} \quad (81)$$

$$= e^{-\lambda n \epsilon} \sum_{k > n(p+\epsilon)} \binom{n}{k} (pe^{\lambda q})^k (qe^{-\lambda p})^{n-k} \quad (82)$$

$$\leq e^{-\lambda n \epsilon} \sum_{k=0}^n \binom{n}{k} (pe^{\lambda q})^k (qe^{-\lambda p})^{n-k} \quad (83)$$

$$= e^{-\lambda n \epsilon} (pe^{\lambda q} + qe^{-\lambda p})^n \quad (84)$$

Using the fact that $e^x \leq x + e^{x^2}$ for all $x \in \mathbb{R}$ ¹, we have

$$\mathbb{P}\left(\frac{N_A}{n} > p + \epsilon\right) \leq e^{-\lambda n \epsilon} (p\lambda q + pe^{\lambda^2 q^2} - q\lambda p + qe^{\lambda^2 p^2})^n \quad (85)$$

$$\leq e^{-\lambda n \epsilon} (pe^{\lambda^2} + qe^{\lambda^2})^n \quad (86)$$

$$= e^{-\lambda n \epsilon + \lambda^2 n}. \quad (87)$$

Since this inequality is true for all $\lambda > 0$, we can minimize the right-hand side by minimizing the exponent $-\lambda n \epsilon + \lambda^2 n$. This is achieved by substituting $\lambda = \frac{\epsilon}{2}$, which gives us

$$\mathbb{P}\left(\frac{N_A}{n} > p + \epsilon\right) \leq e^{-n\epsilon^2/4}, \quad (88)$$

which is also called [Bernstein's inequality](#). We can follow the same steps to obtain

$$\mathbb{P}\left(\frac{N_A}{n} < p - \epsilon\right) \leq e^{-n\epsilon^2/4}. \quad (89)$$

Therefore, we have

$$\mathbb{P}\left(\left|\frac{N_A}{n} - p\right| > \epsilon\right) = \mathbb{P}\left(\frac{N_A}{n} > p + \epsilon\right) + \mathbb{P}\left(\frac{N_A}{n} < p - \epsilon\right) \leq 2e^{-n\epsilon^2/4}. \quad (90)$$

By taking the limit as $n \rightarrow \infty$, we obtain the desired result. $\square$

This proof comes from the proof of Theorem 1 in Section 2.2 of [\[1\]](#).

References

- [1] G. Grimmett and D. Stirzaker, *Probability and random processes*, 3rd ed (Oxford University Press, Oxford ; New York, 2001).
- [2] S. Richard, [Special Mathematics Lecture – Introduction to probability](#), Spring 2026.

¹Consider $f(x) = e^{x^2} + x - e^x$. Since $f'(x) = 2xe^{x^2} + 1 - e^x$ and $f''(x) = 2e^{x^2} + 4x^2e^{x^2} - e^x$, we have $f'(0) = 0$ and $f''(0) = 1 > 0$. Moreover, since $\lim_{x \rightarrow \pm\infty} f(x) = +\infty$, we have $f(x) \geq f(0) = 0$ for all $x \in \mathbb{R}$.